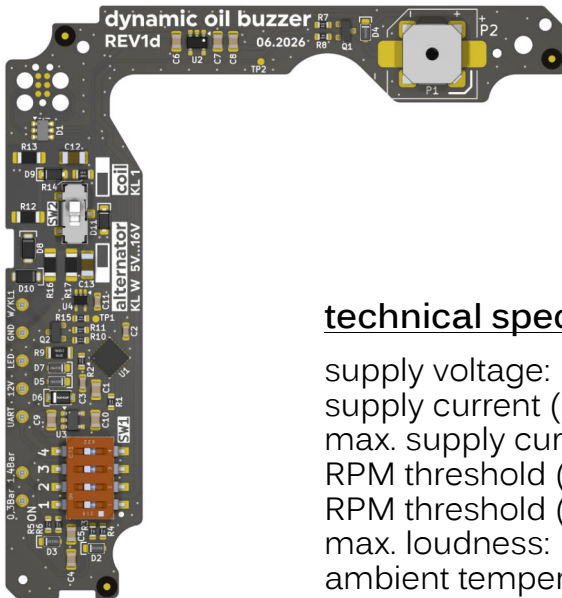


dynamic oil buzzer



github
dynamic oil buzzer

technical specifications

supply voltage:	6V ... 16 V
supply current (idle):	5 mA
max. supply current:	150 mA
RPM threshold (active):	>2050 RPM
RPM threshold (inactive):	<2000 RPM
max. loudness:	~90dBa
ambient temperature:	-40°C ... 85°C

This circuit board serves as a dynamic oil pressure monitor for diesel and benzin/petrol/gasoline engines.

When the engine speed exceeds **2050 RPM**, oil pressure 2 is checked (this function is inactive when the engine speed is below 2000 RPM). Furthermore, the circuit board beeps and flashes if the engine speed is lost despite oil pressure 2 being present (V-belt failure in diesel engines).

If oil pressure 1 is lost, the LED flashes immediately. If another fault occurs, this is signaled after 2 seconds (LED and buzzer).



compatibility:

T3 / vanagon
Golf 1 Cabriolet
Golf 2
Scirocco 2
Caddy 1
Jetta 2
Passat B2
Polo 2



<https://youtu.be/VGgJPHZ6h68>

function overview and
mounting instructions

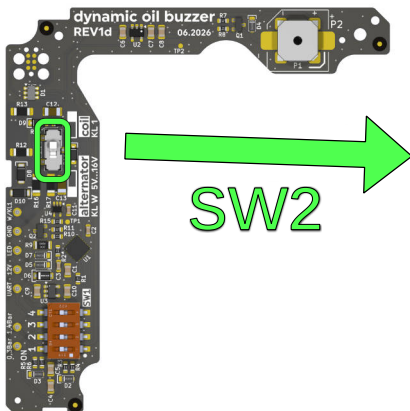
select signal source

The circuit board accepts either a **5V–16V** signal (sine wave, square wave, pulse, etc.) or the **terminal 1** pulse from the ignition coil (Digifant, Digijet, carburetor).

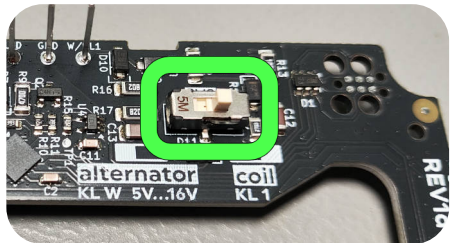
Please note that the switch must be set correctly before installation!



Using the wrong setting can damage the electronics!

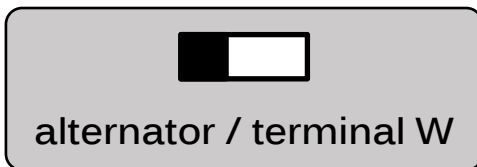


SW2



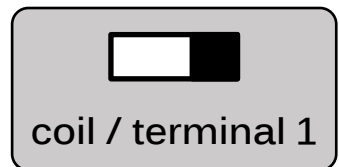
example:
switch position right „COIL“

switch position left



alternator / terminal W

switch position right



coil / terminal 1

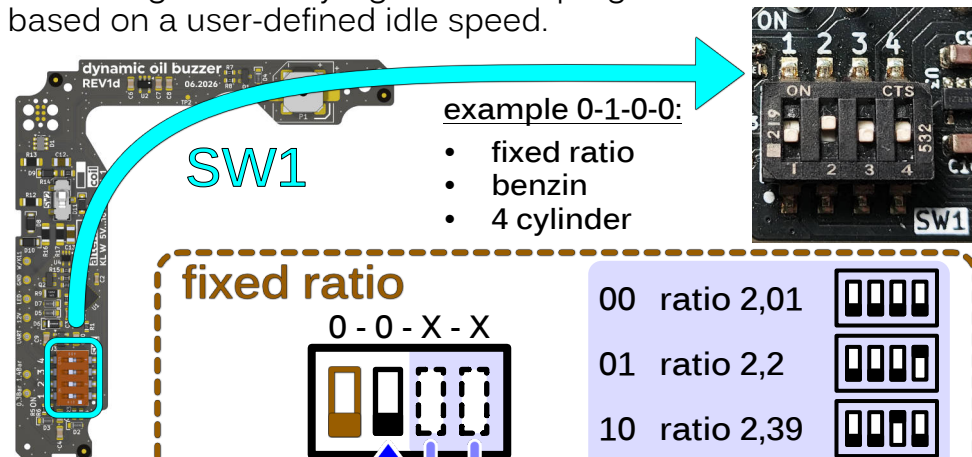
- Alternator
- 5V / 12V square wave
- 5V ... 16V signal
(sine wave, square wave,
pulse, etc.)

- Ignition coil

set the engine type and pulley ratio

A 4-position switch is used to set the engine type, as well as the pulley ratio for diesel engines or the number of cylinders for gasoline engines.

In learning mode, any signal can be programmed and used based on a user-defined idle speed.



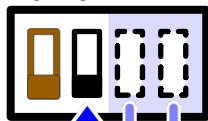
SW1

example 0-1-0-0:

- fixed ratio
- benzin
- 4 cylinder

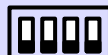
fixed ratio

0 - 0 - X - X



diesel

00 ratio 2,01



01 ratio 2,2



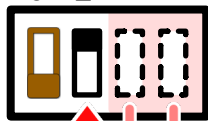
10 ratio 2,39



11 ratio 2,75



0 - 1 - X - X



benzin

00 4 cylinder



01 5 cylinder



10 6 cylinder



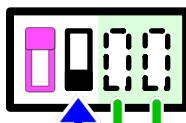
11 8 cylinder



learning
mode

1 - 0 - X - X

diesel



00 750 rpm



01 800 rpm



10 850 rpm

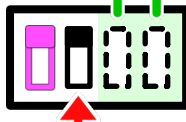


11 900 rpm



1 - 1 - X - X

benzin

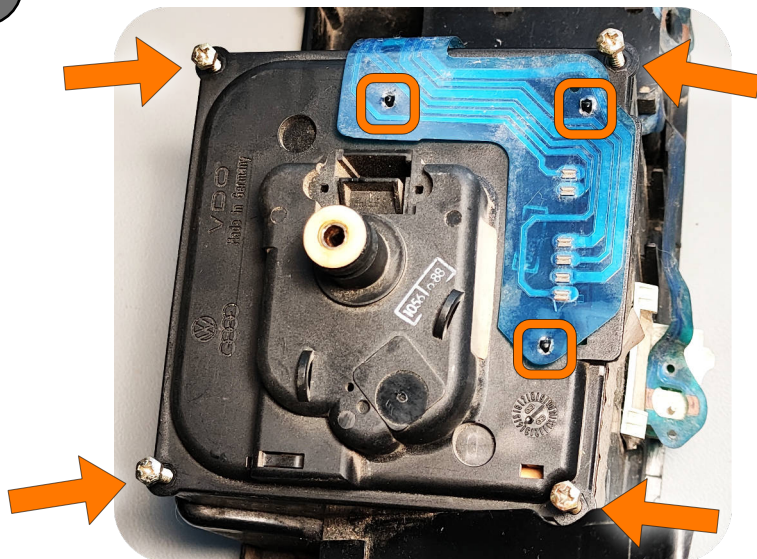


idle-
RPM

Mounting instructions

Once all the DIP switches have been set correctly, the installation can begin.

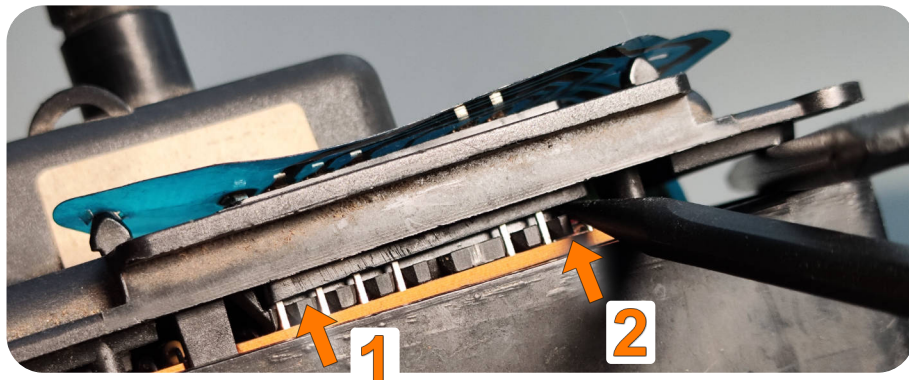
1 Removal of the conductive foil



To remove the connector as carefully as possible, you must first remove all four screws from the speedometer housing. Then carefully unclip the speedometer cover (use a plastic pry bar!).



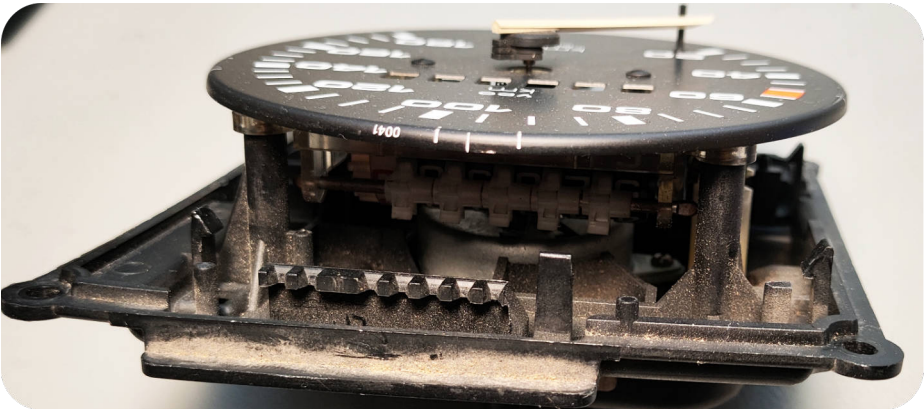
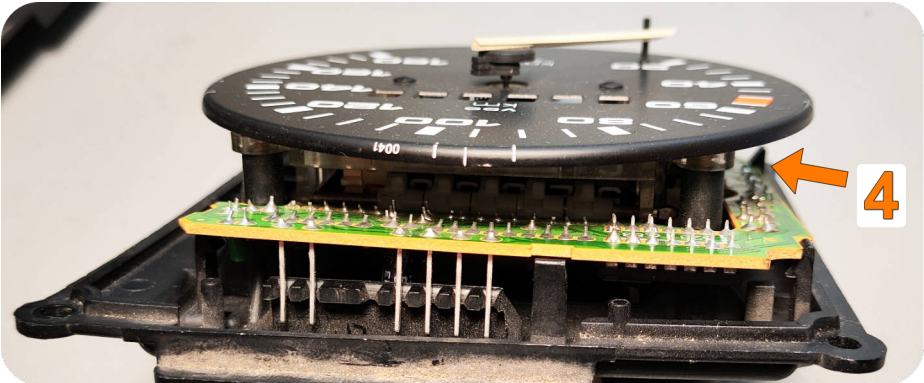
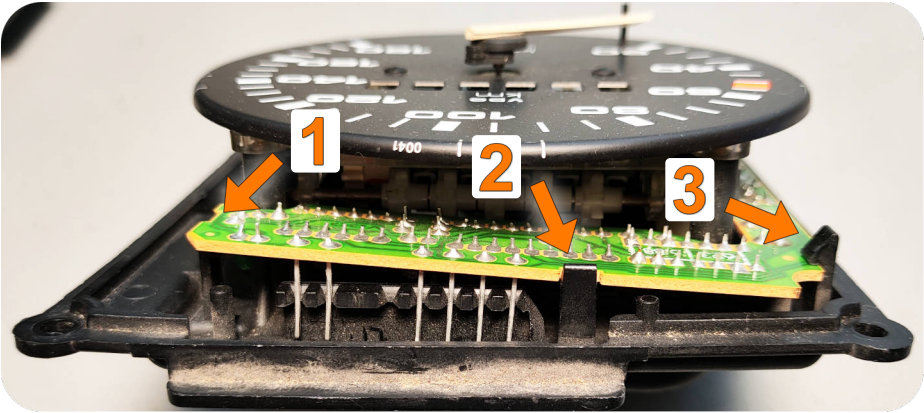
2 Lift the circuit board connector

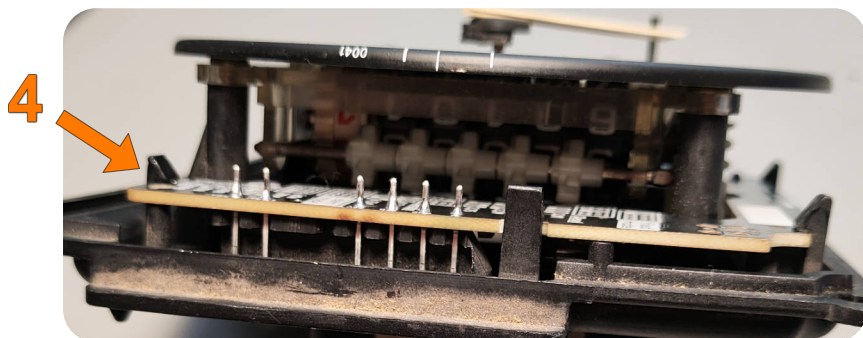
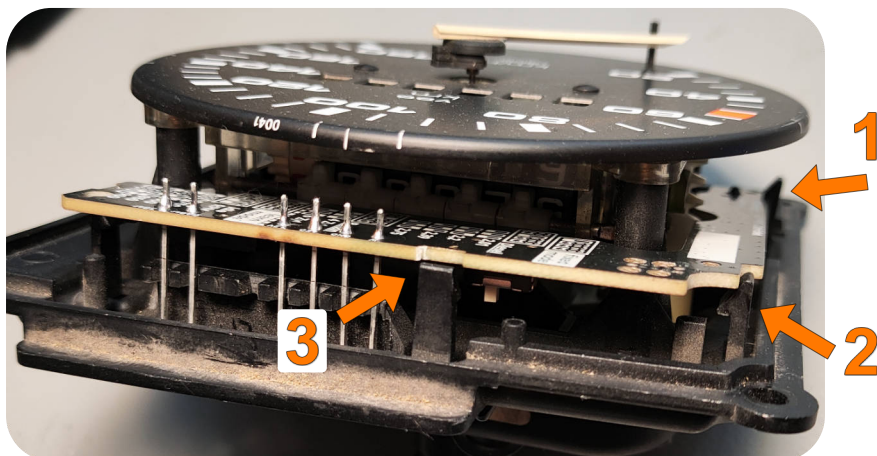
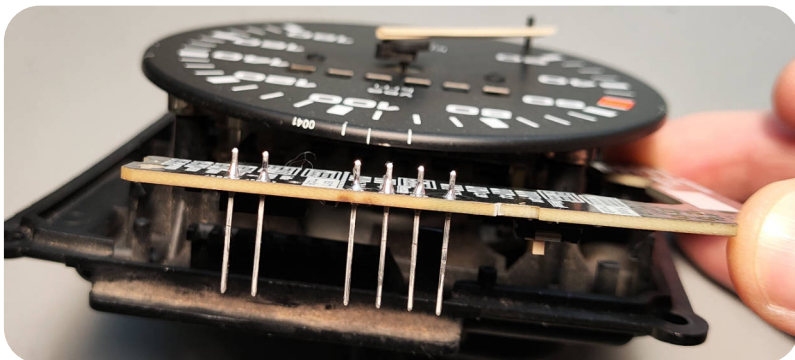


Now lift the plug carefully, alternating between each side.



3 Unclip the circuit board



4**Clip in the new circuit board**

Now screw the speedometer back into the instrument cluster, plug in the connector and fit the instrument cluster.

Commissioning

switch position: **fixed ratio**



Once installed, the car can be operated as normal.

switch position: **learning mode**



To enter learning mode, turn the ignition on 4 times without starting the engine.

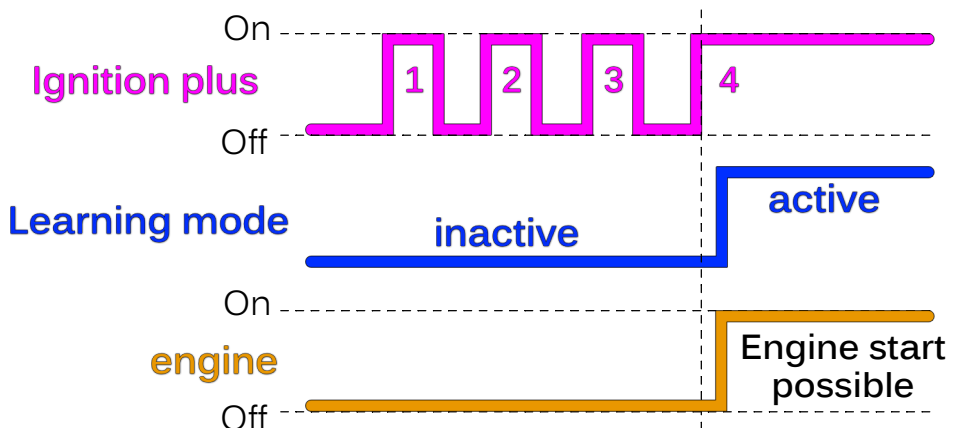
If the engine starts before the 4th attempt, the counter will reset to 0.

As shipped, the learning mode will be executed at first startup when the ignition is switched on.

If the switch position has changed since the last time the unit was powered on, the learning mode will also start the first time the ignition is switched on.

If you wish to enter the learning mode again in the future, you must cycle the ignition on and off 4 times before starting the engine.

Start learning mode



Once the circuit board is in **learning mode**, the buzzer beeps and the LED flashes rhythmically every 1.5 seconds. The engine can now be started and left idling (**Briefly press the throttle, otherwise there will be no RPM signal**).

There is no time limit within which the engine must be started.

Once the engine is started, the buzzer stops only after the RPM signal is detected. The measurement takes place after an **8-second time delay**.

If there is no RPM signal, the buzzer will beep continuously!
Causes: Wiring errors, broken cables, etc...



Once the engine has been started, the idle speed must be kept constant!

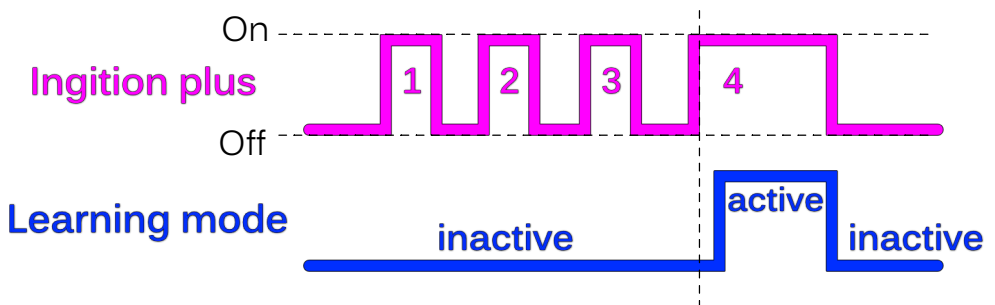


During the measurement, the average is calculated from several readings.

Another series of buzzer sounds, indicating that the current idle speed has been saved.

From now on, the saved values will be used to calculate the correct speed during normal operation.

Abort learning mode

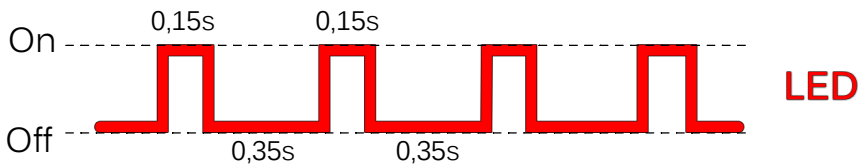


If you wish to abort the learning mode (the buzzer beeps rhythmically every 1.5 seconds), simply switch off the ignition without starting the engine.

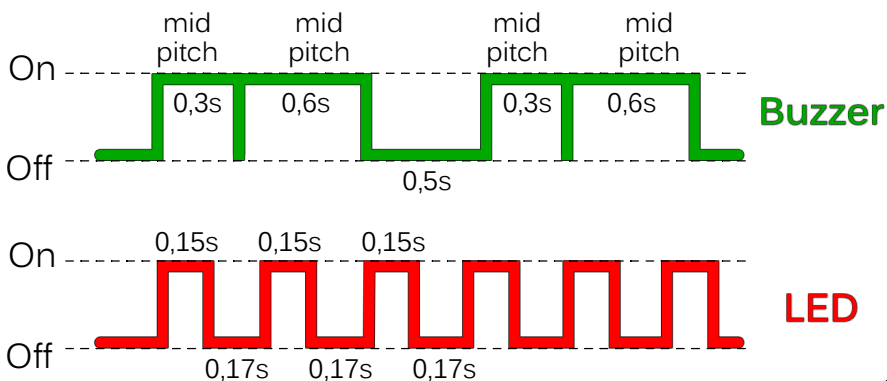
Abortion is only possible if 'previous' values from a learning process have already been saved!

Function overview

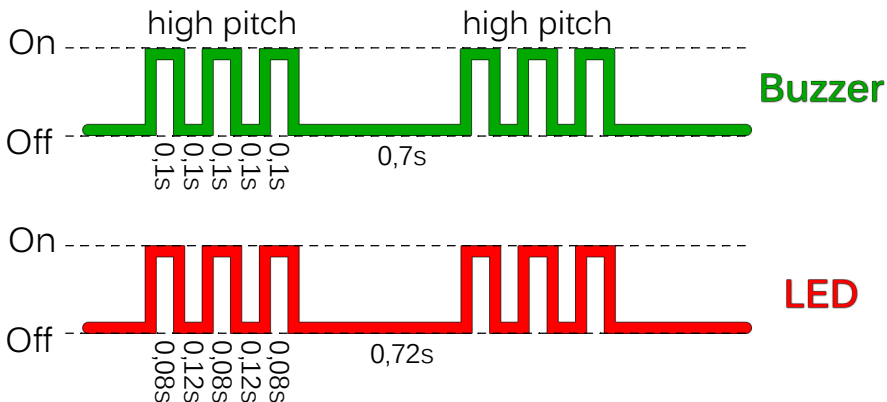
Oil pressure 1 too low



Oil pressure 2 too low and 2050 rpm exceeded



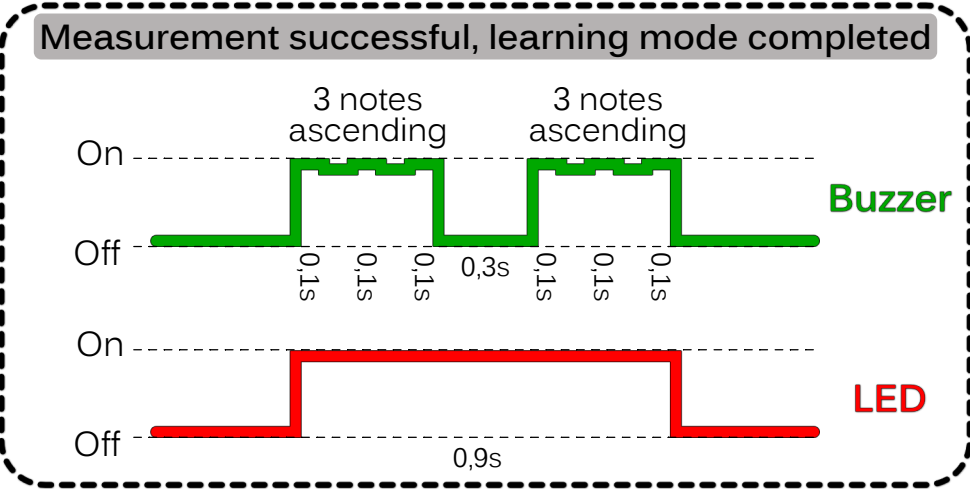
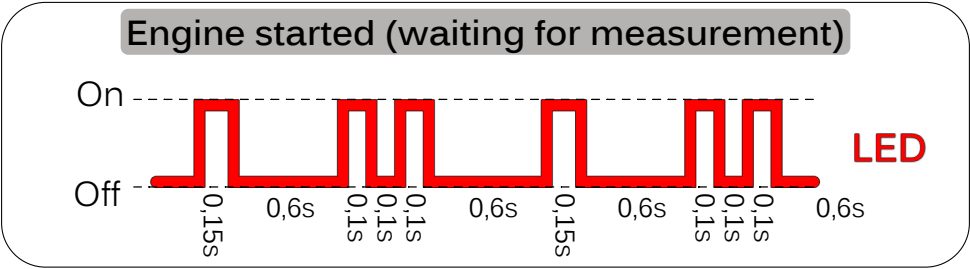
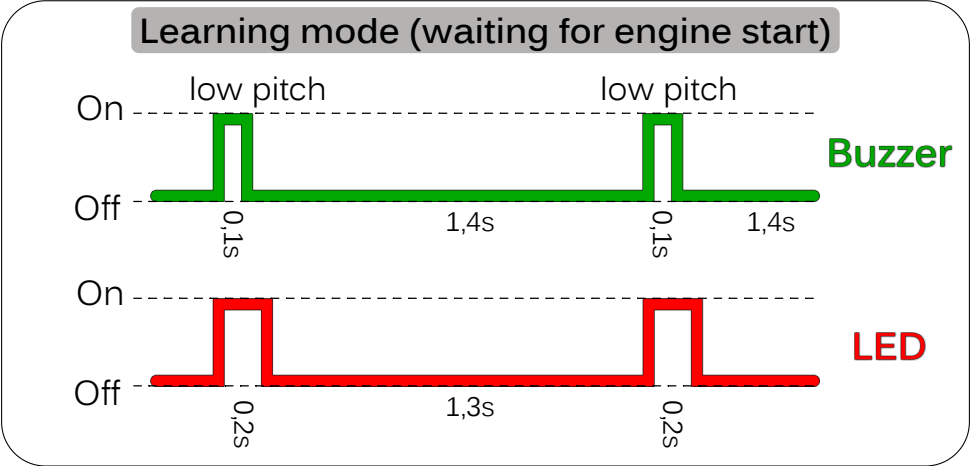
Oil pressure 2 OK / no RPM signal detected



If a speed above 9000 rpm is detected, the buzzer chirps and the LED stays on!



Function overview in learning mode



This sequence is only once and doesn't repeat

Replaces the following dyn. oil pressure boards:

251 919 064 A	251 919 253 F
251 919 064 B	251 919 253 G
251 919 064 C	251 919 253 H
251 919 064 D	251 919 253 J
251 919 064 G	251 919 253 K
321 919 064 M	251 919 253 L

belt ratio (diesel):

- 2,75** 2 belts without power steering (up to 1989)
- 2,39** Diesel with power steering (from 1986)
- 2,2** Diesel without power steering, single belt (from 1989)
- 2,01** Diesel with air conditioning, air conditioning and power steering, 2nd alternator or hydraulic pump

UART Interface

115200 / 8 data / 1 stop / no parity

The 3.3V UART output transmits information on the current RPM, all inputs and outputs, and the fault status every 50 ms.
When the software is started, all stored variables are displayed.

```
COM4 - PuTTY
dynamic oil buzzer REV1c - V20260421
idle RPM 800
SW1 1-0-0-1
EEPROM | PSC   79 | period   802 | idle RPM 900 | learn events   12
EEPROM | SW1 1-0-1-1 | cell active   1 | cell writes 8279

---  learning mode  ---

start engine and keep in idle (800 rpm)
waiting for measurement window (8 seconds)
measurement in progress
---  LEARNING done  ---

EEPROM | PSC   79 | period  2672 | idle RPM 800 | learn events   13

800 rpm | 151 Hz | oil1 ok | oil2 low | BUZZER on | LED on
800 rpm | 151 Hz | oil1 ok | oil2 low | BUZZER on | LED on
800 rpm | 151 Hz | oil1 ok | oil2 low | BUZZER on | LED on
799 rpm | 151 Hz | oil1 ok | oil2 low | BUZZER on | LED on
```